

Instructions to the Students:

1. All questions are compulsory
2. Draw diagrams / figures wherever necessary
3. Figures to right indicate full marks

Q.1. Objective Type Questions (Answer All the Questions) (10X2) = 20

- i) Define microbiology & write scope of pharmaceutical microbiology
- ii) Write the application of animal cell culture in pharmaceutical industry.
- iii) What do you mean by Phase contrast microscopy
- iv) Give difference between bacteriostatic and bactericidal agents.
- v) Classify different methods of sterilization
- vi) Explain clean area classification in short
- vii) What are cell lines
- viii) Give different antimicrobial agents used for preservation of pharmaceutical products
- ix) Define sterility indicator with example.
- x) Mention the factors affecting disinfectant activity

Q.2. Long Answers (Answer 2 out of 3) (10X2) = 20

- i) Define sterilization. Explain the principle, procedure, applications and demerits of sterilization using autoclave
- ii) What are the sources of contamination and their control in aseptic room
- iii) Classify bacteria on the basis of nutritional requirements and add a note on raw materials used for Preparation of culture media.

Q.3. Short Answers (Answer 7 out of 9) (5 X 7) = 35

- i) Describe bacterial growth curve
- ii) Write a note on cultivation of viruses
- iii) Differentiate between Total count and viable count
- iv) Explain assay of Vitamin B12
- v) Describe construction and working of laminar air flow
- vi) Discuss various methods of evaluation of bacteriostatic disinfectant
- vii) Enlist types of spoilage. Explain different factors affecting microbial spoilage of Pharmaceuticals
- viii) Write about Gram's & Acid fast staining.
- ix) Classify fungi & give the pharmaceutical uses of fungi

*** END OF THE PAPER ***